

DS	Session	Queue	Question Information	Level 1	Challenge 51
Problem Description The Monk recently learnt about priority queues and requested his teacher for a fun challenge to solve. As a result, his teacher devised a simple task. He now possesses A, which is an integer array. He wishes to discover the product of the greatest, second largest, and third largest integers in the range $[1, i]$ for each index i. Note: Two numbers can be the same value-wise but they should be distinct index-wise. Constraints: $1 \leq N \leq 100000$ $0 \leq A[i] \leq 1000000$ Input: The first line contains an integer N, denoting the number of elements in the array A. The next line contains N space separated integers, each denoting the ith integer of the array A. Output: Print the answer for each index in each line. If there is no second largest or third largest number in the array A upto that index, then print "-1", without the quotes.					

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
1 2 3 4 5

EXPECTED OUTPUT

-1
-1
6
24
60

Test Case 2

INPUT (STDIN)

8
3 7 1 2 4 7 5 9

EXPECTED OUTPUT

-1
-1
21
42
84
196
245
441

▼ Mandatory Test Cases

Test Case 1

KEYWORD

if(biggest<big)

Test Case 2

KEYWORD

if(a[i]>biggest)

▼ Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

10

TOKEN COUNT

300

NLOC

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